

AMEEMAH HUMAYUN

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RESEARCH INTERESTS

Culturally situated human-AI interaction in low-resource and health contexts, with a focus on language, literacy, and social norms; LLM safety and evaluation, spanning cross-lingual, multi-turn, adversarial, and expert-validated benchmarking, automated model testing, and statistical analysis of failure modes; and sociotechnical HCI, particularly privacy and shared-device access, AI mental models, and the ways AI redistributes healthcare and information-seeking work.

EDUCATION

Lahore University of Management Sciences (LUMS)	2022 - 2026
BS Computer Science	Lahore, Pakistan
Cambridge International O & A Levels – Beaconhouse Liberty Campus	2017 - 2022
SAT: 1550; A Levels: 1A*, 3A; O Levels: 8A*, 2A	Lahore, Pakistan

PUBLICATIONS

- A. Humayun, B. Zubair, and M. Mustafa. *Beyond Euphemisms: Rethinking LLMs for SRH in Conservative Contexts*. ACM CHI '26, Barcelona.
- S. Shanza, A. Humayun, U. Hengartner, and L. Zhang-Kennedy. *Sharing Digital Devices is Normal and Cultural: Privacy and Security Challenges in Collectivist Immigrant Households*. USENIX SOUPS '26, Hannover.
- (Under Review) A. Humayun, B. Zubair, and M. Mustafa. *Patient Safety Implications of LLM-Mediated Fetal Sex Disclosure in Conservative Healthcare Contexts*. Nature Health, 2026.

AWARDS, SERVICE & TEACHING

Gary Marsden Travel Award, SIGCHI	February 2026
Student Volunteer, CHI 2026, Barcelona	April 2026
Teaching Assistant, CS 5606: Designing Human-Centered AI, LUMS	Jan - June 2026

RESEARCH & WORK EXPERIENCE

Awaaz-e-Sehat & The National AI Hub— UX Researcher & Designer	July 2026 - Present
Maternal-health AI, clinical workflows & evaluation — PI: Dr. Maryam Mustafa	Lahore, Pakistan

- Lead methods design for a **2-site, 500-patient comparative maternal-health pilot**, building **7 linked instruments** that integrate survey, usability, workflow-timing, consultation-observation, provider, and facility-level data to evaluate the effects of **AI-generated EMR summaries** on care quality and clinical workflows.
- Led iterative redesign of the **electronic medical record (EMR)**, reducing intake time by **78% (90 → 20 min)** while retaining **~70–75% of clinically relevant information**, and translating **35+ maternal and obstetric risk criteria** into automated backend flags.
- Redesigned the **AI-generated clinical summary** for rapid provider review, restructuring key obstetric variables and risk information for **10–20-second interpretation** during high-volume consultations.
- Design and conduct usability research with patients and frontline health workers, examining **comprehension, workflow fit, low-literacy interaction, and shared or mediated phone access**, and translate findings into product iterations.
- Led design and development of a **low-literacy maternal-health prototype for traditional birth attendants (TBAs)**, in collaboration with **Aga Khan University and VITAL Pakistan**.
- Synthesized **official TBA curricula, WHO guidance, and focus-group findings from frontline TBAs** into the prototype's information architecture, content structure, and interaction flows; iteratively refined the design around **information overload, literacy constraints, usability, and referral needs in Pakistan**.
- Supervised three qualitative research projects examining (1) **how low-literacy, digitally excluded users form mental models of technically opaque AI systems**; (2) **how women turn to GenAI following dismissive clinical encounters**, shaping subsequent information-seeking and care trajectories; and (3) a **multi-country study with Google on how women perceive AI personhood** and how these perceptions shape trust and engagement.
- Conduct field visits across partner healthcare sites and design patient-facing recruitment and onboarding materials, including **clinic handout cards and Facebook ads**, iterating messaging, visual hierarchy, and accessibility for **low-literacy audiences**.

Safe Interactions Lab, University of Waterloo — Research Assistant	Nov 2025 - June 2026
Shared-device privacy in collectivist immigrant households (USENIX SOUPS '26) — Advisors: Dr. Leah Zhang-Kennedy, Dr. Urs Hengartner	Remote

- Co-led qualitative research on **privacy and security in shared-device use** across **18 South Asian immigrant families (41 participants)**.
- Served as one of **two primary qualitative coders**; analyzed **~50% of interviews** in Atlas.ti using reflexive thematic analysis and co-developed and iteratively refined the study codebook.
- Synthesized coded data across research questions on **sociocultural drivers, privacy/security risks, and unmet design needs**, contributing to the study's broader framing of **shared devices as socially and culturally governed infrastructures**.
- Co-developed: (1) a **5-category taxonomy** of shared-device privacy and security risks; and (2) a **desires-to-needs framework** translating participants' stated feature requests into underlying **privacy, access-control, and autonomy needs**.

- Substantially drafted and revised the **findings and discussion**, translating empirical insights into design recommendations for **multi-user authentication, identity-aware access, and configurable privacy controls**.

Interactive Media Lab, LUMS — Research Project Lead

May 2025 – June 2026

Conversational AI safety and fetal sex disclosure in healthcare (Nature Health, under review) — Advisor: Dr. Maryam Mustafa

Lahore, Pakistan

- Led mixed-methods research on safety risks from LLM-mediated fetal-sex disclosure, in collaboration with **Awaaz-e-Sehat**.
- Designed the interview protocol; recruited and interviewed 13 OB/GYNs and radiologists across two hospital sites, and coordinated clinical annotation of study materials.
- Conducted inductive thematic analysis in ATLAS.ti, iteratively developing the qualitative codebook and synthesizing practitioner accounts into a **four-pathway clinical harm framework**.
- Translated qualitative findings into a locally grounded **clinical reference standard** for evaluating model disclosure behavior and the conceptual basis for a technical audit of conversational AI safety.
- Designed a **3-condition adversarial evaluation** spanning single-, 6-, and 18-turn conversations to test guardrail degradation under sustained conversational pressure, surfacing failures missed by single-turn benchmarks, including hallucinated disclosure and over-refusal.
- Built a Python-based multi-provider LLM evaluation pipeline spanning 3 model providers and 3 language conditions; engineered resumable multimodal API orchestration and automated statistical analysis for disclosure-rate and self-contradiction metrics.
- Led manuscript drafting and contributed to the study's research framing, analysis, and interpretation.

Interactive Media Lab, LUMS — Research Project Lead

May 2025 – June 2026

LLM performance on conservative, indirect health language (CHI '26) — Advisor: Dr. Maryam Mustafa

Lahore, Pakistan

- Led mixed-methods research on culturally situated LLM failures in maternal and sexual/reproductive health communication in Pakistan; **accepted to CHI 2026 (CORE A*)** and presented in Barcelona. Project associated with **Awaaz-e-Sehat** and supported by the Gates Foundation.
- Designed and conducted qualitative fieldwork comprising **40 clinician–patient observations, 22 semi-structured interviews, and 5 focus groups** with patients and healthcare practitioners.
- Adapted the study methodology in response to early field findings, shifting from primarily elicited accounts toward naturalistic observation and clinician interviews to better capture embedded euphemistic and indirect language.
- Developed through bottom-up qualitative analysis: (1) a novel **Domain–Approach framework** for classifying indirect health communication by referential domain and communicative strategy; and (2) a decision-tree classification procedure applied across ~95 patient-derived expressions.
- Co-designed and executed a systematic evaluation of **5 LLMs on 71 culturally grounded Roman Urdu utterances** across controlled prompting conditions, with gynecologist validation and statistical model comparison; demonstrated how strong aggregate scores can mask clinically and contextually important failures, and characterized recurrent failure modes including polysemy blindness, script instability, and persona inconsistency.
- Synthesized qualitative and model-evaluation findings into design recommendations for culturally situated health AI, including terminology management, miscommunication-aware interaction design, and privacy-conscious multimodal input.

Other UX & Interaction Design Projects

2024

Introductory projects in UX research, interaction design, and prototyping

Lahore, Pakistan

- 45+ interviews and usability studies across *Rehnuma* (accessibility ratings, w/ PSRD), *Safarnama* (cultural tourism), *TurboTrack* (booking system); user research, design, prototyping, testing. Tools: *Figma, HTML/CSS, Atlas.ti*.

SKILLS & INTERESTS

Research Methods: Semi-structured and expert interviews, focus groups, contextual inquiry, usability testing, reflexive thematic analysis, inductive coding, codebook development

LLM Evaluation: Single- and multi-turn evaluation (naturalistic and adversarial); cross-lingual LLM benchmarking (English/Roman Urdu) across OpenAI, Anthropic, and Google models; human and expert evaluation; failure-mode, safety, and self-contradiction analysis; statistical model comparison with exact binomial confidence intervals (Clopper–Pearson)

Research & Design Tools: ATLAS.ti, MAXQDA, Dedoose, Figma, Miro, Zotero, LaTeX/Overleaf, Canva, Procreate, Blender, Nomad Sculpt

Programming & Technical: Python, C/C++, JavaScript/React, HTML/CSS, MATLAB, Git

Languages: English, Urdu, Punjabi, Hindi